FortuneCloud Technologies

MCA Internship Program

PROJECT

EC2 Instance Launch with Enforced Tagging Policy

Using AWS Tag Policies & IAM

Student Name	Adnan
Email	adnanparbhulkar06@gmail.com
Phone	9322169982
Location	Pune

1. Introduction

Cloud cost management is a critical challenge for enterprise organizations. Without proper resource tagging, it becomes impossible to track resource ownership, allocate costs accurately, and enforce governance policies.

This project demonstrates how to enforce mandatory tagging on AWS EC2 instances to ensure every virtual machine is properly identified and attributed. The project implements tag enforcement using AWS IAM Policies and validates behavior by attempting instance launches both with and without the required tags.

2. Objective

The goal of this project is to:

- Understand and apply AWS tagging best practices.
- Configure a tag enforcement policy using AWS IAM.
- Launch an EC2 instance successfully when all mandatory tags are provided.
- Demonstrate that launching an EC2 instance without required tags is denied.
- Document all steps with screenshots for verification.

3. Required Tags Definition

As per the Cloud Governance mandate, every EC2 instance must have the following four tags at the time of launch:

Tag Key	Tag Value (Used in This Project)
Name	Adnan
emailID	adnanparbhulkar06@gmail.com
phoneNo	9322169982
Place	Pune

These tags are used for identification, contact tracing, and resource grouping across the AWS Organization.

4. Tag Policy Implementation — IAM Deny Policy

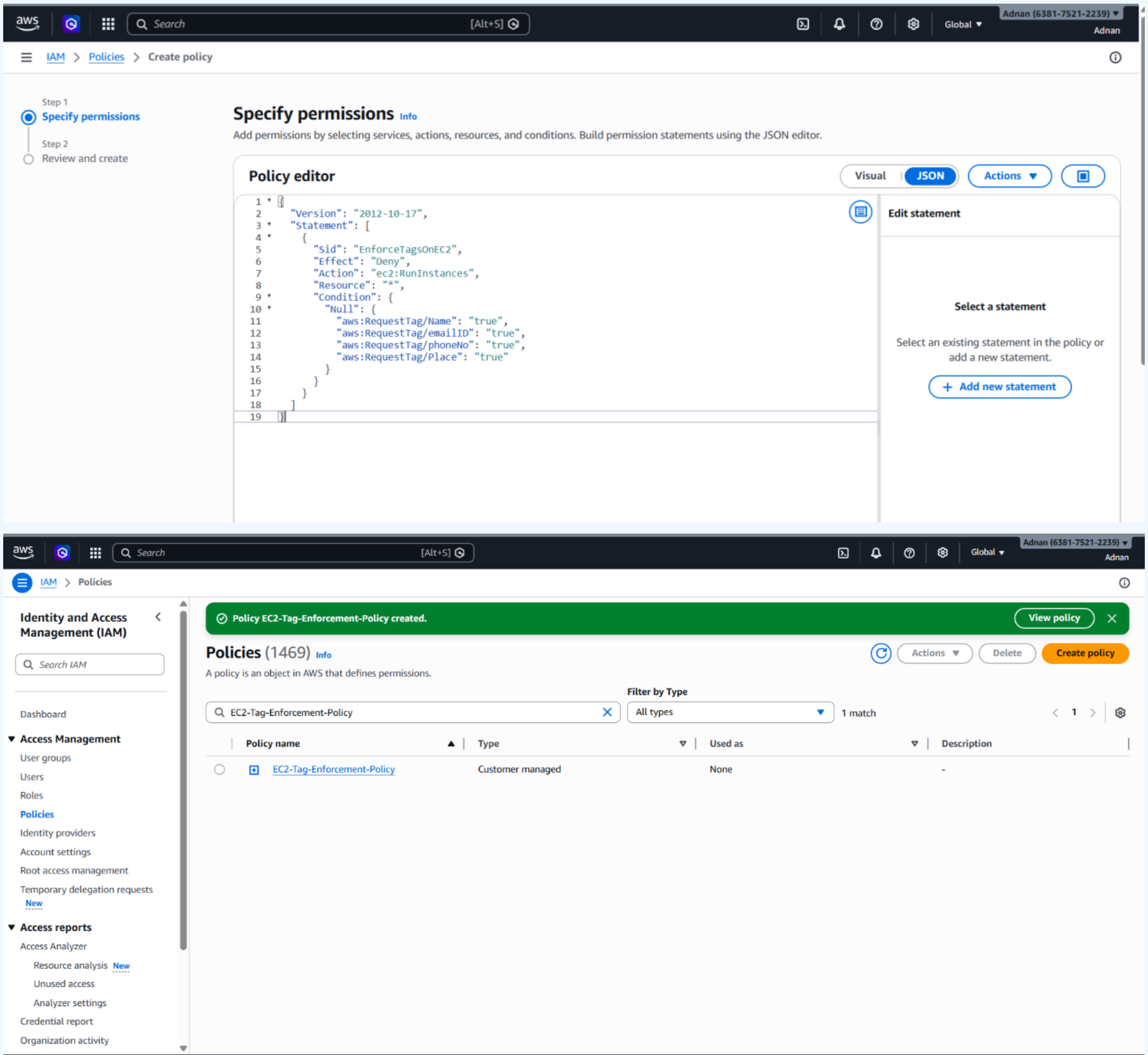
4.1 Policy Description

An IAM Deny Policy was created to block the ec2:RunInstances action whenever any of the four mandatory tags are missing. This policy uses the Condition key aws:RequestTag with a Null check to detect missing tags.

4.2 IAM Policy JSON

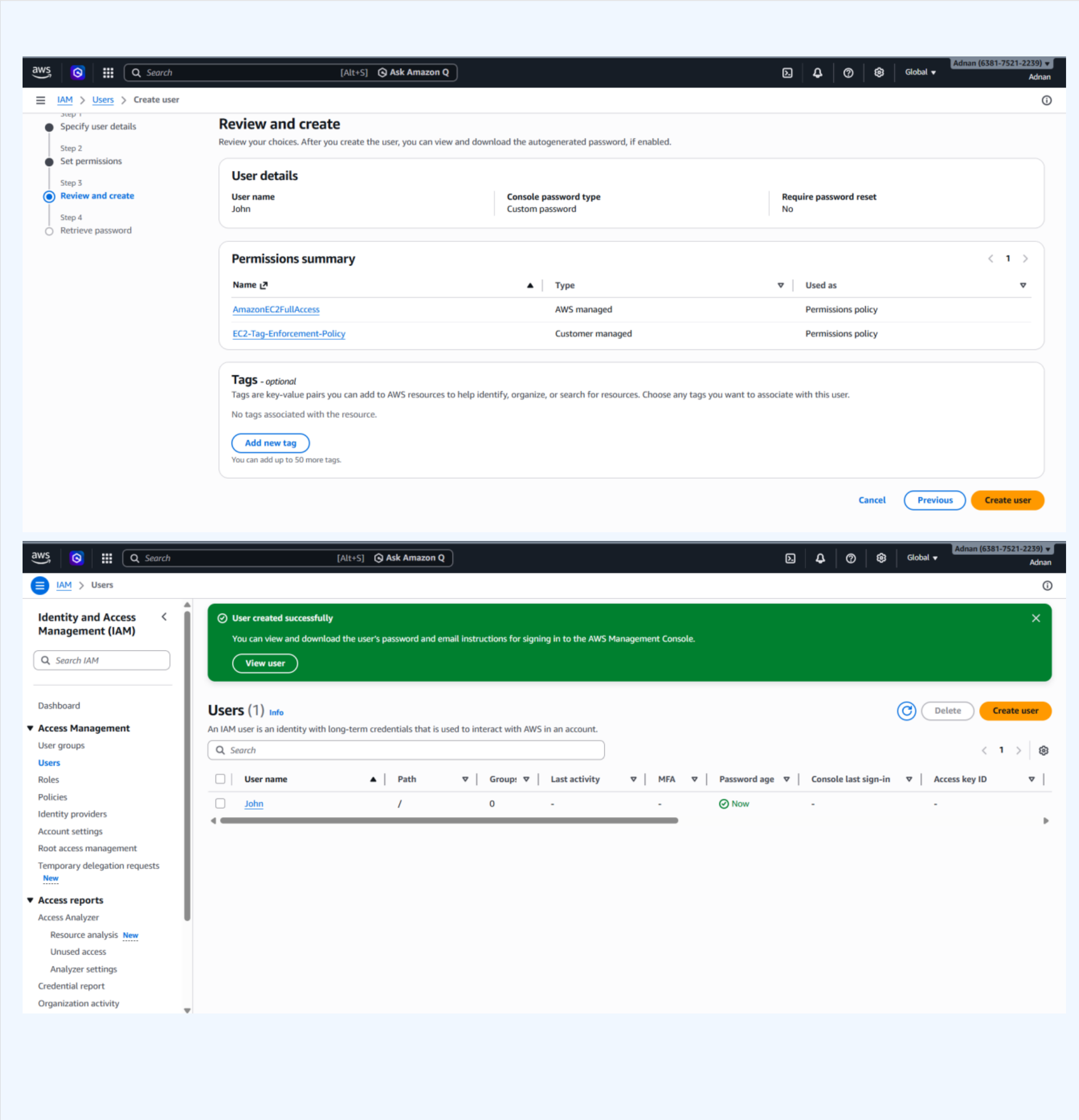
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyEC2WithoutRequiredTags",
      "Effect": "Deny",
      "Action": "ec2:RunInstances",
      "Resource": "arn:aws:ec2:*:*:instance/*",
      "Condition": {
        "Null": {
          "aws:RequestTag/Name": "true",
          "aws:RequestTag/emailID": "true",
          "aws:RequestTag/phoneNo": "true",
          "aws:RequestTag/Place": "true"
        }
      }
    }
  ]
}
```

4.3 Screenshot — IAM Policy Creation



4.4 Policy Attached to IAM User/Role

The deny policy was attached to the IAM user used for testing EC2 launches, along with the AmazonEC2FullAccess managed policy.



5. EC2 Launch — WITHOUT Tags (Failure Test)

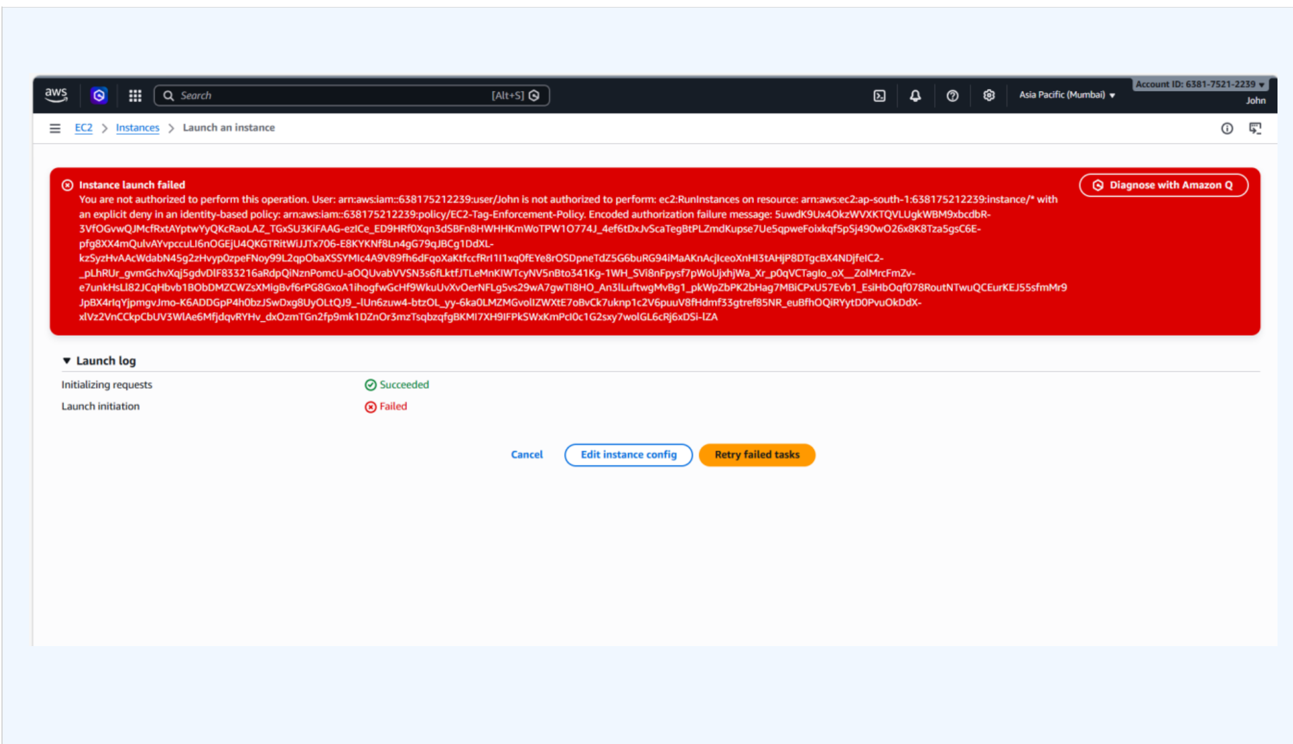
5.1 Description

To validate that the tag enforcement policy works correctly, an EC2 instance launch was attempted without providing the mandatory tags. The expected result is an authorization error denying the action.

5.2 Expected Result

- Action DENIED by IAM policy.
- Error message: 'User is not authorized to perform ec2:RunInstances'.
- No instance is created in the EC2 dashboard.

5.3 Screenshot — Launch Failure (Error Message)



6. EC2 Launch — WITH All Required Tags (Success)

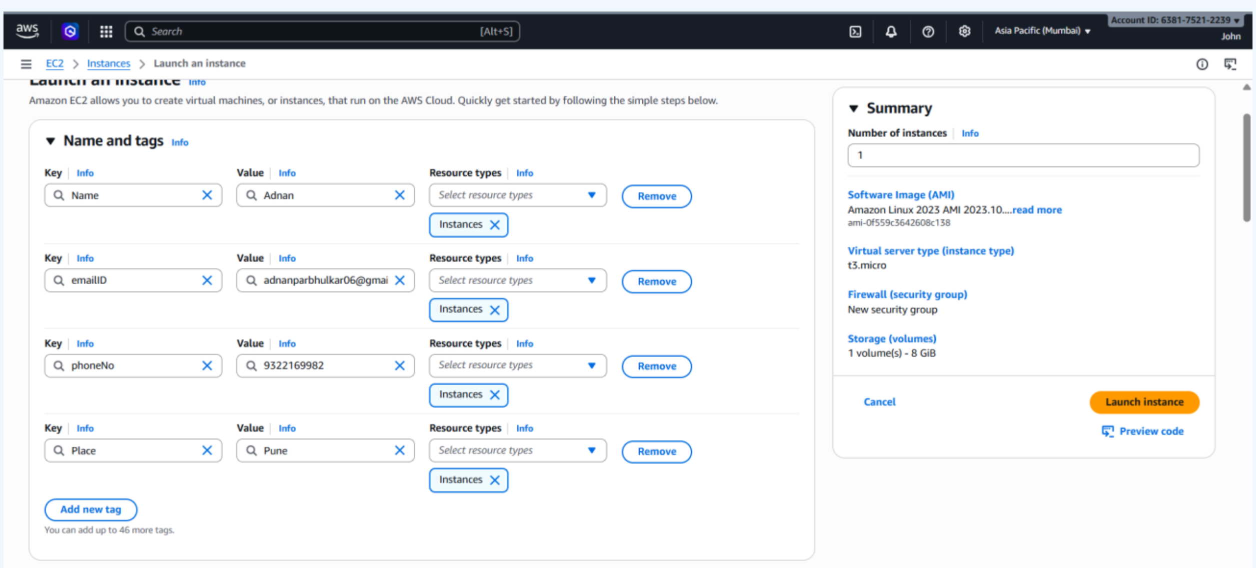
6.1 Description

An EC2 instance was launched with all four mandatory tags provided. The launch was successful, confirming that the IAM policy correctly allows tagged instances and denies untagged ones.

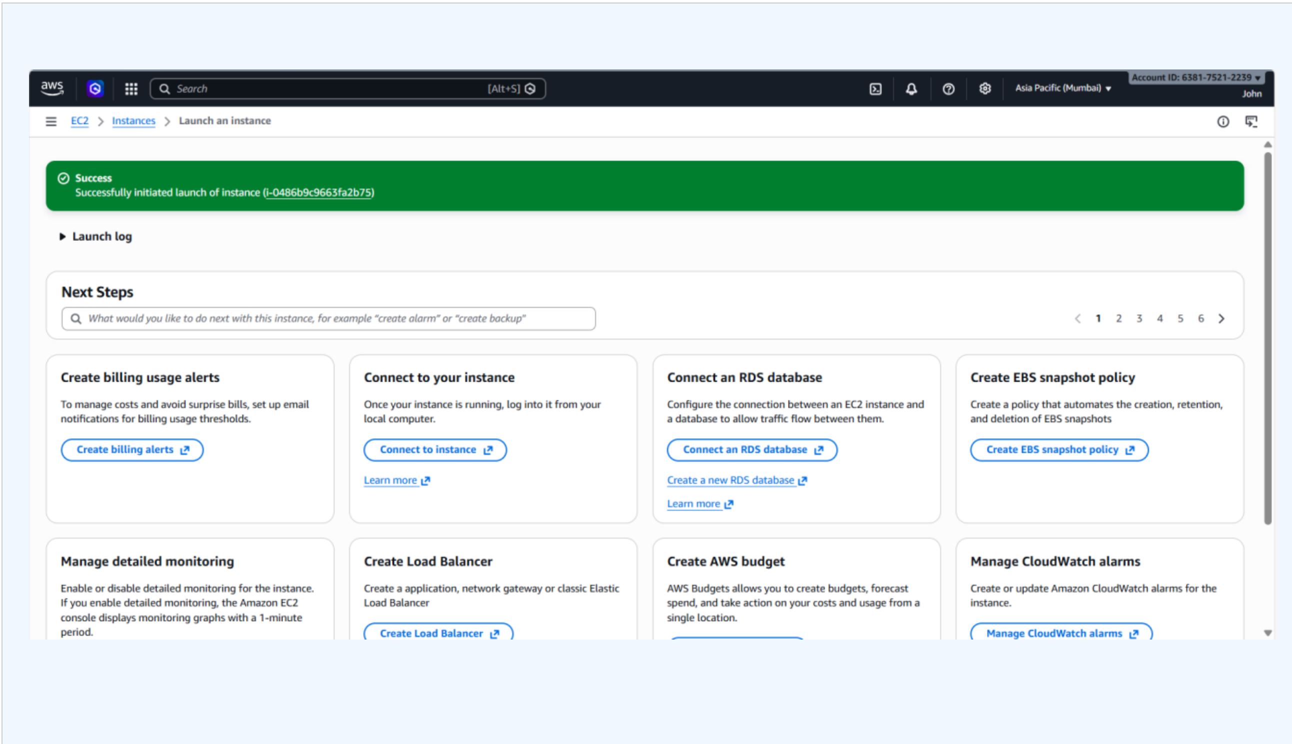
6.2 Tags Applied at Launch

Tag Key	Value Applied
Name	Adnan
emailID	adnanparbhulkar06@gmail.com
phoneNo	9322169982
Place	Pune

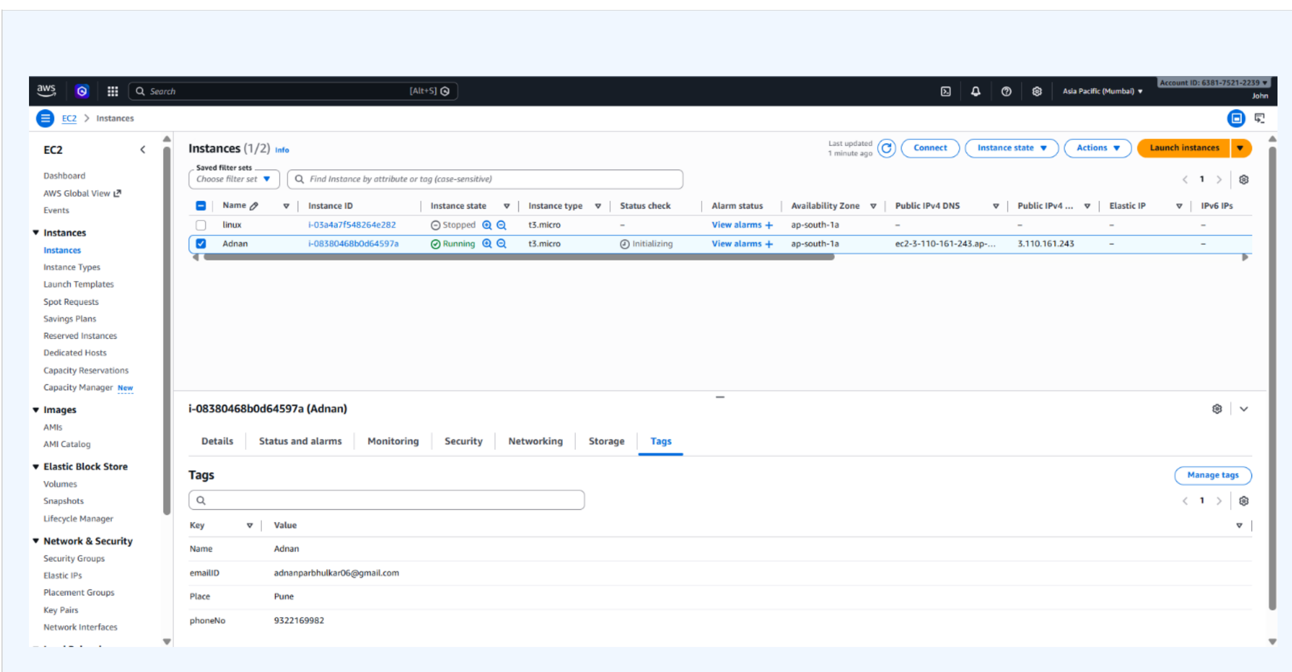
6.3 Screenshot — Tags Added in AWS Console



6.4 Screenshot — Successful Instance Launch



6.5 Screenshot — Tags Visible on Running Instance



7. Step-by-Step Execution Guide

Step 1 — Log in to AWS Console

- Go to https://console.aws.amazon.com
- Log in with your IAM or root credentials.
- Select region: ap-south-1 (Mumbai).

Step 2 — Create the IAM Deny Policy

- Navigate to IAM > Policies > Create Policy.
- Click the JSON tab and paste the deny policy JSON.
- Name the policy: EC2-Tag-Enforcement-Policy.
- Click Create Policy.

Step 3 — Attach Policy to IAM User

- Go to IAM > Users > Select your test user.
- Click Add Permissions > Attach Policies Directly.
- Attach both AmazonEC2FullAccess and EC2-Tag-Enforcement-Policy.

Step 4 — Test Launch WITHOUT Tags (Failure)

- Go to EC2 > Instances > Launch Instances.
- Fill in all settings but skip adding any tags.
- Click Launch Instance — observe the authorization error.
- Take a screenshot of the error message.

Step 5 — Launch WITH All Required Tags (Success)

- Go to EC2 > Instances > Launch Instances.
- Under Name and Tags, add all 4 tags (Name, emailID, phoneNo, Place).
- Set Resource Type to Instances for each tag.
- Configure: Amazon Linux 2023 AMI, t3.micro, default security group.
- Click Launch Instance — instance should launch successfully.
- Take screenshots of tag configuration and running instance.

Step 6 — Verify Tags on Running Instance

- Go to EC2 > Instances > select your instance.
- Click the Tags tab to confirm all 4 tags are present.
- Take a screenshot for documentation.

Step 7 — Terminate the Instance

- Right-click the instance > Terminate Instance.
- This avoids any unwanted AWS billing charges.

8. Results Summary

Test Scenario	Expected Result	Actual Result
Launch WITHOUT tags	**DENIED**	Authorization error — Instance not created
Launch WITH all 4 tags	**SUCCESS**	Instance launched and running successfully

9. Conclusion

This project successfully demonstrated how AWS IAM Deny Policies can be used to enforce mandatory resource tagging for EC2 instances. By requiring the Name, emailID, phoneNo, and Place tags at the time of launch, the organization ensures:

- Every EC2 instance is traceable to an individual.
- Cost allocation and billing can be done accurately per user.
- Cloud governance policies are enforced automatically without manual intervention.
- Untagged resources are blocked before they incur costs.

This approach can be scaled across the AWS Organization using Service Control Policies (SCPs) via AWS Organizations for stronger enforcement at the account level.

10. References

- AWS Documentation — IAM Policies: https://docs.aws.amazon.com/IAM/latest/UserGuide/
- AWS Documentation — EC2 Tagging: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Using_Tags.html
- AWS Documentation — Tag Policies: https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_tag-policies.html
- FortuneCloud Technologies — MCA Internship Guidelines